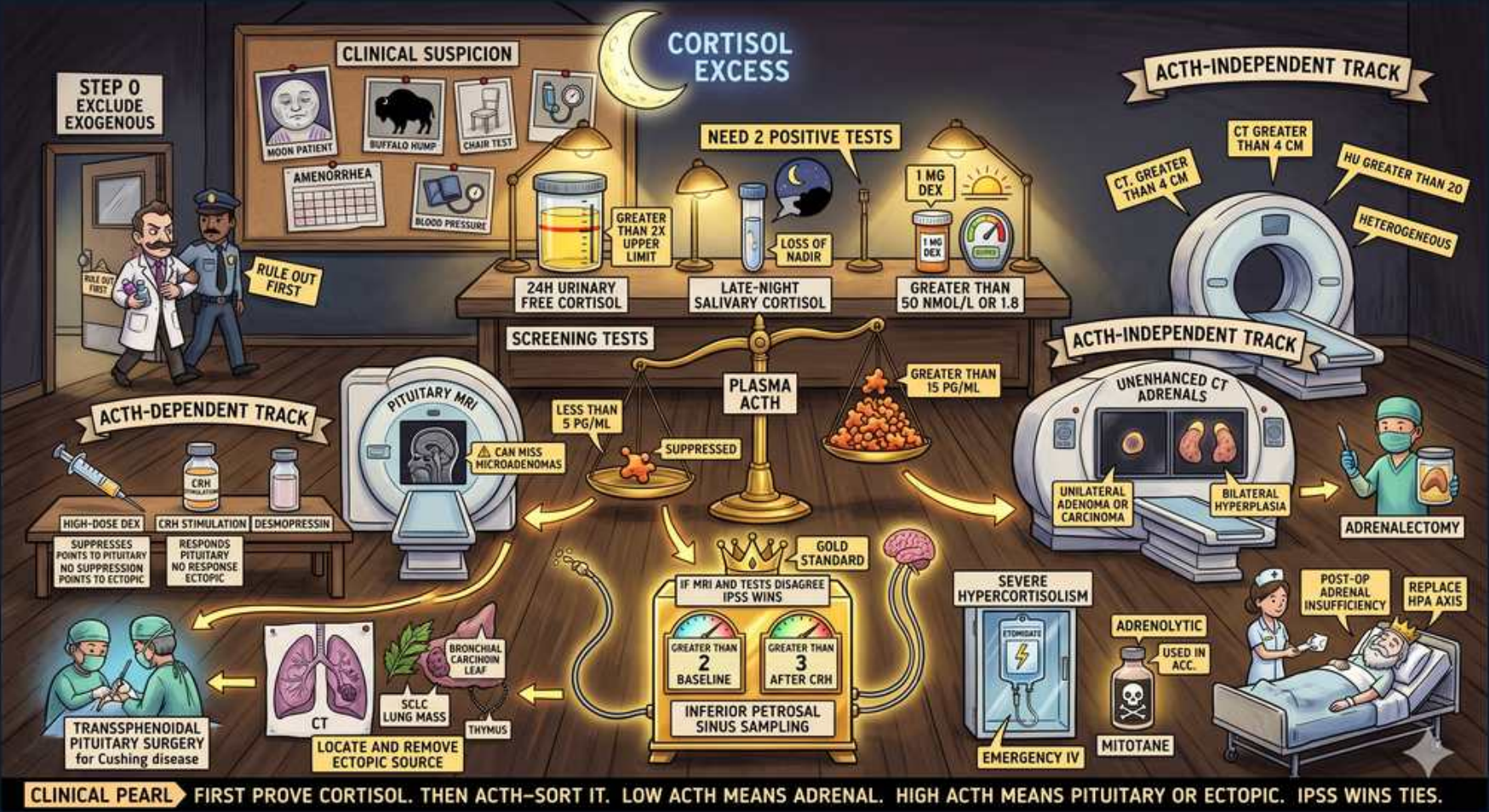


Cushing Syndrome — Confirm Cortisol, Then ACTH-Sort



1. Step 0: exclude exogenous steroids

WHAT YOU SEE

Step 0 board: rule out exogenous glucocorticoids first

WHAT IT ENCODES

Before any testing, exclude iatrogenic Cushing from exogenous glucocorticoids — the most common cause overall. Exogenous steroids give a Cushingoid body with LOW ACTH and LOW endogenous cortisol.

BOARD TRAP

Most common Cushing cause is exogenous steroids; suppressed ACTH AND low cortisol distinguish it from disease.

2. Need 2 positive screening tests

WHAT YOU SEE

Banner: need 2 positive tests to confirm cortisol excess

WHAT IT ENCODES

Confirm hypercortisolism with at least two abnormal first-line tests before localizing: 24h urinary free cortisol, late-night salivary cortisol, and/or 1 mg overnight DST. One positive test alone is insufficient.

BOARD TRAP

Don't localize off a single abnormal test — require two concordant positives to confirm Cushing.

3. 24h urinary free cortisol

WHAT YOU SEE

Jug: 24h UFC greater than 3x upper limit

WHAT IT ENCODES

Elevated 24-hour urinary free cortisol (often >3–4x ULN) reflects integrated daily cortisol production and is a core screening test. Collection errors and high fluid intake can confound.

BOARD TRAP

A markedly high UFC (>3x ULN) is strong evidence; mild elevations need a second confirmatory test.

4. Late-night salivary cortisol

WHAT YOU SEE

Moon icon: late-night salivary cortisol, loss of nadir

WHAT IT ENCODES

Cushing abolishes the normal nighttime cortisol nadir, so an elevated late-night salivary cortisol is sensitive. Two collections increase reliability.

BOARD TRAP

Loss of the late-night cortisol nadir is the hallmark; a normal late-night value argues against Cushing.

5. 1 mg overnight DST

WHAT YOU SEE

Pill: 1 mg dex, cortisol >50 nmol/L or 1.8 fails to suppress

WHAT IT ENCODES

After 1 mg dexamethasone at 11 pm, morning cortisol >1.8 µg/dL (50 nmol/L) fails to suppress and screens positive. False positives occur with estrogen, CYP inducers, and depression.

BOARD TRAP

Cortisol >1.8 after low-dose dex = fails to suppress = abnormal; estrogen/CBG can cause false positives.

6. Plasma ACTH sorts the cause

WHAT YOU SEE

Center: plasma ACTH is the gold standard branch point

WHAT IT ENCODES

Once hypercortisolism is confirmed, measure plasma ACTH. LOW/suppressed ACTH (<5 pg/mL) = ACTH-independent (adrenal). HIGH/normal ACTH = ACTH-dependent (pituitary or ectopic).

BOARD TRAP

ACTH direction is the master sort: low = adrenal source, high = pituitary/ectopic.

7. ACTH < 5 → adrenal (independent)

WHAT YOU SEE

Branch: ACTH less than 5, suppressed

WHAT IT ENCODES

Suppressed ACTH points to a primary adrenal source — adenoma, carcinoma, or bilateral hyperplasia. Proceed to adrenal CT, not pituitary MRI.

BOARD TRAP

Low ACTH means go to adrenal imaging; ordering a pituitary MRI here is the wrong next step.

8. ACTH > 15 → pituitary/ectopic

WHAT YOU SEE

Branch: ACTH greater than 15 pg/mL, dependent

WHAT IT ENCODES

Elevated/normal ACTH means an ACTH-dependent cause. Next distinguish pituitary (Cushing disease, ~80%) from ectopic ACTH (small-cell lung, carcinoid) using high-dose dex, CRH, and imaging.

BOARD TRAP

High ACTH still requires separating pituitary from ectopic — don't stop at 'ACTH-dependent.'

9. High-dose dex test

WHAT YOU SEE

ACTH-dependent track: high-dose dexamethasone, CRH stimulation

WHAT IT ENCODES

High-dose dex suppresses cortisol in most pituitary tumors (retain feedback) but NOT in ectopic sources. CRH stimulation raises ACTH/cortisol in pituitary disease, not ectopic.

BOARD TRAP

Pituitary tumors suppress with high-dose dex and respond to CRH; ectopic tumors do neither.

10. Pituitary MRI

WHAT YOU SEE

Pituitary MRI machine in ACTH-dependent track

WHAT IT ENCODES

Pituitary MRI is obtained when ACTH is high and dynamic testing suggests Cushing disease. Microadenomas <6 mm may be invisible, so a negative MRI does not exclude pituitary source.

BOARD TRAP

A normal pituitary MRI does NOT rule out Cushing disease — proceed to IPSS when MRI is small/negative.

11. IPSS — gold standard

WHAT YOU SEE

Inferior petrosal sinus sampling, central/peripheral gradient before and after CRH

WHAT IT ENCODES

When MRI and dynamic tests disagree, inferior petrosal sinus sampling settles pituitary vs ectopic. A central-to-peripheral ACTH gradient (≥ 2 baseline, ≥ 3 after CRH) confirms pituitary source. IPSS wins ties.

BOARD TRAP

IPSS is the tiebreaker — a positive central gradient confirms pituitary even with a negative MRI.

12. Ectopic ACTH — find the tumor

WHAT YOU SEE

SCLC/carcinoid imaging, locate and remove ectopic source

WHAT IT ENCODES

Ectopic ACTH (small-cell lung cancer, bronchial carcinoid, MTC) gives very high ACTH/cortisol, severe hypokalemia, and rapid onset. Localize with chest/abdomen CT and somatostatin-receptor imaging.

BOARD TRAP

Severe hypokalemic alkalosis + rapid Cushing + smoker = ectopic ACTH (SCLC), not pituitary disease.

13. ACTH-independent → adrenal CT

WHAT YOU SEE

Right track: unenhanced adrenal CT, unilateral adenoma vs carcinoma vs bilateral hyperplasia

WHAT IT ENCODES

With suppressed ACTH, image the adrenals: unilateral adenoma (lipid-rich, low HU) vs adrenocortical carcinoma (large, irregular, high HU) vs bilateral macronodular/PPNAD hyperplasia. Treatment is adrenalectomy.

BOARD TRAP

Low-ACTH Cushing goes to adrenal CT; size/HU separate benign adenoma from carcinoma.

14. Adrenalectomy → transient AI

WHAT YOU SEE

Post-op adrenal insufficiency, needs HPA axis support

WHAT IT ENCODES

After removing a cortisol-secreting adrenal tumor, the suppressed contralateral adrenal/HPA axis cannot respond, so post-operative adrenal insufficiency is expected — give stress-dose glucocorticoid replacement and taper.

BOARD TRAP

Expect post-adrenalectomy adrenal insufficiency from a chronically suppressed axis — cover with steroids, don't withhold.

15. Clinical pearl banner

WHAT YOU SEE

Banner: prove cortisol, then ACTH-sort; low=adrenal, high=pituitary/ectopic, IPSS wins

WHAT IT ENCODES

Pearl: first PROVE hypercortisolism (two tests), then ACTH-SORT. Low ACTH = adrenal; high ACTH = pituitary vs ectopic, resolved by high-dose dex/CRH and ultimately IPSS.

BOARD TRAP

The whole algorithm: confirm cortisol first, measure ACTH second; never localize before confirming and ACTH-sorting.